

Closing the Tsirelson Loophole in Relativistic Quantum Field Theory: Split Inclusions, Hyperfinite Local Algebras, and the Effective Dimension of the Vacuum

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Abstract

MIP*=RE implies that correlations of commuting observables on an infinite-dimensional Hilbert space can lie at a finite distance from every limit of finite-dimensional tensor-product correlations. Such correlations would let a finite experiment detect an actual infinity with a modest number of rounds. In earlier work we identified this as the only route that escapes the resource bounds on certifying Hilbert-space dimension. Here we ask whether relativistic quantum field theory allows Bell correlations in this gap.

The answer is no under standard structural assumptions. If the theory has the split property, which follows from the Buchholz–Wichmann nuclearity condition and holds for free fields, then the correlations of local measurements in strictly spacelike separated regions are tensor-product correlations and lie in C_{qa} . If the local algebras are hyperfinite, as Buchholz, D’Antoni and Fredenhagen showed under a scaling-limit assumption, the same holds for regions that are merely spacelike, with no gap between them. We give a self-contained proof of the second statement from semidiscreteness. Alice’s measurements are compressed through matrix algebras by unital completely positive maps, and the resulting correlations converge to the original ones. The static-patch algebra of de Sitter space constructed by Chandrasekaran, Longo, Penington and Witten is a corner of a crossed product of such an algebra by $\mathbb{R}$, so it is again injective. Thus, in these frameworks, the gap found by MIP*=RE cannot be realised by local measurements.

We then bound how much dimension a Bell test can certify from the state side. Compressing each party to the support of its ε -smoothed reduced state reproduces all correlations to within $2(\sqrt{\varepsilon_A} + \sqrt{\varepsilon_B}) + \varepsilon_A + \varepsilon_B$. An experiment with N rounds can therefore certify at most the smooth max-entropy of the local state at smoothing $\varepsilon \sim 1/N$. For the vacuum of a lattice free scalar field we compute this effective dimension exactly. It follows an area law, at most 9 bits for a 128-site block at $\varepsilon = 10^{-6}$ against 553 bits from naive counting. It grows only from 2.3 to 13.1 bits as ε decreases from 10^{-1} to 10^{-12} , with a local slope that falls from 0.60 to 0.19 bits per bit of precision. A product bound over thermal modes shows that the growth is polylogarithmic in $1/\varepsilon$ for any regularised block.

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1 Introduction

Bell experiments are usually modelled in two ways. In the tensor-product model, Alice and Bob act on the factors of $\mathcal{H}_A \otimes \mathcal{H}_B$. In the commuting-operator model, which is the natural framework of algebraic quantum field theory [1], their observables act on a single Hilbert space and merely commute. Let C_q denote the correlations of finite-dimensional tensor-product strategies, C_{qa} its closure, C_{qs} the correlations of possibly infinite-dimensional tensor-product strategies, and C_{qc} the commuting-operator correlations. Then $C_q \subseteq C_{qs} \subseteq C_{qa} \subseteq C_{qc}$ [2, 3]. Tsirelson’s problem asked whether $C_{qa} = C_{qc}$. It is equivalent to Connes’ embedding problem [4, 3], and $\text{MIP}^* = \text{RE}$ answered both in the negative [5].

The answer matters for the question of whether an actual infinity is empirically accessible. In a companion paper [6] we showed that certifying a Hilbert-space dimension beyond holographic bounds with known dimension witnesses requires more experimental rounds than the region can perform. The single exception is the gap between C_{qa} and C_{qc} . A correlation in $C_{qc} \setminus C_{qa}$ has a finite distance from every finite-dimensional model, so a finite number of rounds could exhibit it. Cabello, Quintino and Kleinmann listed four logical possibilities for the physical status of such correlations, ranging from “never realisable” to “realisable by spacelike separated parties” [7]. They mention the split property and the role of nuclear algebras, but they do not draw a conclusion for quantum field theory.

We show that standard structural results of algebraic QFT settle the question for local measurements.

- *Split inclusions* (Theorem 1). If Alice’s region has a neighbourhood that is still spacelike to Bob’s region, the split property makes the correlations tensor-product correlations. They lie in $C_{qs} \subseteq C_{qa}$.
- *Hyperfiniteness* (Theorem 2). If Alice’s local von Neumann algebra is injective, equivalently hyperfinite, every correlation with commuting partners lies in C_{qa} , even without a gap between the regions.
- *de Sitter space* (Proposition 3.1). The type II_1 algebra of an observer in a de Sitter static patch is injective under the same assumptions.
- *Effective dimension* (Section 4). An N -round Bell test certifies at most the smooth max-entropy of the local state at smoothing $\varepsilon \sim 1/N$. For the vacuum of a lattice free field this is small, follows an area law and grows polylogarithmically in $1/\varepsilon$.

2 Correlation sets and local algebras

A Bell scenario has finitely many settings x, y and outcomes a, b . A commuting-operator strategy consists of a Hilbert space $\mathcal{H}$, a normal state ω on $B(\mathcal{H})$, and POVMs $\{A_a^x\}, \{B_b^y\}$ with $[A_a^x, B_b^y] = 0$. The correlation is $p(a, b|x, y) = \omega(A_a^x B_b^y)$. We use two standard facts.

Lemma 2.1 ([2, 3]). $C_{qs} \subseteq C_{qa}$: correlations of tensor-product strategies on infinite-dimensional spaces are limits of finite-dimensional ones.

Lemma 2.2. If Alice’s POVM elements lie in a finite-dimensional C^* -algebra F , and Bob’s commute with F , then the correlation lies in C_{qs} .

Proof. A unital representation of $F \cong \bigoplus_k M_{n_k}$ decomposes $\mathcal{H} = \bigoplus_k \mathbb{C}^{n_k} \otimes \mathcal{K}_k$, with F acting as $\bigoplus_k X_k \otimes 1$ and with commutant $\bigoplus_k 1 \otimes B(\mathcal{K}_k)$. Hence Alice’s operators have the form

$\bigoplus_k X_k \otimes 1$ and Bob's the form $\bigoplus_k 1 \otimes Y_k$. The subspace $\bigoplus_k \mathbb{C}^{n_k} \otimes \mathcal{K}_k$ embeds isometrically into $(\bigoplus_k \mathbb{C}^{n_k}) \otimes (\bigoplus_k \mathcal{K}_k)$, where $\bigoplus_k X_k$ and $\bigoplus_k Y_k$ act on the two factors and agree with the original operators on the image. The state is supported on the image, so the correlation is unchanged and lies in C_{qs} . $\square$

Local quantum physics. A net $O \mapsto \mathcal{A}(O)$ assigns von Neumann algebras on the vacuum Hilbert space to regions of Minkowski space, with locality, $\mathcal{A}(O_1) \subseteq \mathcal{A}(O_2)'$ for spacelike separated regions [1]. A local measurement in O is a POVM with elements in $\mathcal{A}(O)$. Two structural properties are known to hold in large classes of theories.

- *Split property.* For $O \Subset \tilde{O}$ (the closure of O lies in the interior of $\tilde{O}$) there is a type I factor $\mathcal{N}$ with $\mathcal{A}(O) \subseteq \mathcal{N} \subseteq \mathcal{A}(\tilde{O})$ [8]. It follows from the Buchholz–Wichmann nuclearity condition, which bounds the energy-level density of localised states, and holds for free fields [9].
- *Hyperfiniteness.* Under the standard postulates and the existence of a scaling limit, each local algebra is isomorphic to $\mathcal{R} \otimes \mathcal{Z}$, where $\mathcal{R}$ is the unique hyperfinite factor of type III₁ [10] and $\mathcal{Z}$ is the centre [11]. In particular local algebras are injective.

3 Local measurements cannot reach the Tsirelson gap

Theorem 1 (Split inclusions). *Let $O_A \Subset \tilde{O}_A$ and let O_B be spacelike to $\tilde{O}_A$. If the split property holds for $O_A \subseteq \tilde{O}_A$, then for every normal state and all local measurements in O_A and O_B the correlation lies in $C_{qs} \subseteq C_{qa}$.*

Proof. Let $\mathcal{N}$ be a type I factor with $\mathcal{A}(O_A) \subseteq \mathcal{N} \subseteq \mathcal{A}(\tilde{O}_A)$. Locality gives $\mathcal{A}(O_B) \subseteq \mathcal{A}(\tilde{O}_A)'\subseteq \mathcal{N}'$. A type I factor induces a tensor factorisation $\mathcal{H} \cong \mathcal{K}_1 \otimes \mathcal{K}_2$ with $\mathcal{N} = B(\mathcal{K}_1) \otimes 1$ and $\mathcal{N}' = 1 \otimes B(\mathcal{K}_2)$. Alice's POVM elements lie in $B(\mathcal{K}_1) \otimes 1$, Bob's in $1 \otimes B(\mathcal{K}_2)$, and the state is a density operator on $\mathcal{K}_1 \otimes \mathcal{K}_2$. The correlation is therefore in C_{qs} , and Lemma 2.1 gives C_{qa} . $\square$

For regions without a gap, for example complementary wedges, a type I factor between them does not exist. Injectivity replaces it. Recall that a von Neumann algebra $\mathcal{M}$ is *semidiscrete* if there are nets of normal unital completely positive (ucp) maps $\alpha_k : \mathcal{M} \rightarrow M_{n_k}$ and $\beta_k : M_{n_k} \rightarrow \mathcal{M}$ with $\beta_k \circ \alpha_k \rightarrow \text{id}_{\mathcal{M}}$ pointwise in the σ -weak topology. Semidiscreteness is equivalent to injectivity and to hyperfiniteness [12, 13].

Theorem 2 (Injective local algebras). *Let $\mathcal{M} \subseteq B(\mathcal{H})$ be an injective von Neumann algebra, ω a normal state, $\{A_a^x\} \subseteq \mathcal{M}$ Alice's POVMs and $\{B_b^y\} \subseteq \mathcal{M}'$ Bob's POVMs. Then $p(a, b|x, y) = \omega(A_a^x B_b^y)$ lies in C_{qa} .*

Proof. Take α_k, β_k as above. Since M_{n_k} is finite-dimensional, hence nuclear, the algebraic tensor product $M_{n_k} \odot \mathcal{M}'$ carries a unique C^* -norm, and we write $M_{n_k} \otimes \mathcal{M}'$ for its completion. Since $\beta_k : M_{n_k} \rightarrow \mathcal{M}$ is completely positive and its range commutes with $\mathcal{M}'$, the map $X \otimes Y \mapsto \beta_k(X)Y$ extends to a completely positive map on $M_{n_k} \otimes \mathcal{M}'$ [14]. Hence $\varphi_k(X \otimes Y) = \omega(\beta_k(X)Y)$ is a state on $M_{n_k} \otimes \mathcal{M}'$. The operators $\tilde{A}_a^x = \alpha_k(A_a^x)$ form POVMs in M_{n_k} because α_k is ucp. Consider the strategy given by the GNS representation of φ_k , with Alice measuring $\tilde{A}_a^x$ and Bob measuring B_b^y . Its correlation is

$$p_k(a, b|x, y) = \varphi_k(\tilde{A}_a^x \otimes B_b^y) = \omega(\beta_k(\alpha_k(A_a^x)) B_b^y). \quad (1)$$

Alice's algebra in this representation is the image of the finite-dimensional algebra M_{n_k} , so $p_k \in C_{qs}$ by Lemma 2.2. As $k \rightarrow \infty$, $\beta_k \circ \alpha_k(A_a^x) \rightarrow A_a^x$ σ -weakly. Since $Z \mapsto \omega(ZB_b^y)$ is a normal functional, $p_k \rightarrow p$. Thus p lies in the closure of C_{qs} , which is C_{qa} by Lemma 2.1. $\square$

Remark. Theorem 2 is the von Neumann-algebraic form of the known fact that C_{qa} and C_{qc} agree when one party's algebra is nuclear [2, 7]. It is also closely related to the tensor-product characterisations of semidiscreteness by Effros and Lance [13]. We include the short proof because the compression maps α_k make the finite-dimensional approximation explicit.

Corollary 3.1. *In a local net with hyperfinite local algebras, every Bell correlation between local measurements in spacelike separated regions O_A, O_B lies in C_{qa} . If only the split property is known, the same holds whenever O_A has a neighbourhood spacelike to O_B . In either case no Bell functional separating C_{qa} from C_{qc} can be violated by local measurements in spacelike separated regions. Among the four possibilities of [7], such theories therefore exclude the fourth, in which correlations in $C_{qc} \setminus C_{qa}$ are realised by local observers in spacelike separated regions. They are silent on the second and third, which involve measurements on a single system or without spacelike separation.*

Proof. Locality gives $\mathcal{A}(O_B) \subseteq \mathcal{A}(O_A)'$. Apply Theorem 2 with $\mathcal{M} = \mathcal{A}(O_A)$, or Theorem 1. $\square$

de Sitter space. Chandrasekaran, Longo, Penington and Witten constructed the algebra of observables accessible to an observer in a static patch of de Sitter space, gravitationally dressed to the observer's worldline [15]. It is a von Neumann algebra of type II_1 , obtained by projecting the crossed product of the static-patch QFT algebra by its modular flow. It has a maximum-entropy state whose entropy agrees, up to a constant, with the generalised entropy.

Proposition 3.1. *If the static-patch QFT algebra is injective, as in the hyperfinite case above, then the de Sitter observer algebra of [15] is injective. Consequently all correlations between this observer and any party whose observables commute with the observer's lie in C_{qa} .*

Proof. Crossed products of injective von Neumann algebras by actions of amenable locally compact groups, such as $\mathbb{R}$, are injective, and so are corners $p\mathcal{M}p$ of injective algebras [16, 12]. Apply Theorem 2. $\square$

4 The effective dimension certified by a Bell test

Theorems 1 and 2 say that local QFT correlations can be approximated by finite-dimensional ones. The next question is how large the approximating dimension must be at a given precision. For a density operator ρ , let $d_\rho(\varepsilon)$ be the smallest rank of a projector P with $\text{tr}(P\rho) \geq 1 - \varepsilon$, so that $\log_2 d_\rho(\varepsilon)$ is the smooth max-entropy $H_0^\varepsilon(\rho)$ [17].

Proposition 4.1. *Let ρ_{AB} be a state on $\mathcal{H}_A \otimes \mathcal{H}_B$ with marginals ρ_A, ρ_B , and let $\varepsilon_A, \varepsilon_B \in (0, 1)$. There is a state supported on a subspace of local dimensions $d_{\rho_A}(\varepsilon_A)$ and $d_{\rho_B}(\varepsilon_B)$ whose correlations, for all POVMs, differ from those of ρ_{AB} by at most*

$$\delta = 2(\sqrt{\varepsilon_A} + \sqrt{\varepsilon_B}) + \varepsilon_A + \varepsilon_B \quad (2)$$

in total variation, for every pair of settings. Consequently, a test that certifies local dimension larger than $\max(d_{\rho_A}(\varepsilon_A), d_{\rho_B}(\varepsilon_B))$ with error $\leq \delta_0$ against ρ_{AB} needs $N \gtrsim \ln(1/4\delta_0)/\delta^2$ rounds.

Proof. Let P_A, P_B be the optimal projectors. By the gentle-measurement lemma [18], $\|\rho_{AB} - (P_A \otimes 1)\rho_{AB}(P_A \otimes 1)\|_1 \leq 2\sqrt{\varepsilon_A}$, and similarly for B . Normalising $\sigma = (P_A \otimes P_B)\rho_{AB}(P_A \otimes P_B)$ costs at most $\varepsilon_A + \varepsilon_B$ in trace norm. On σ , a POVM $\{A_a\}$ acts as the compressed POVM $\{P_A A_a P_A\}$ on $P_A \mathcal{H}_A$. Trace distance bounds the total-variation distance of all outcome distributions. The bound on N follows from Theorem 1 of [6] with ε_G replaced by δ . $\square$

The smoothing parameter enters quadratically. For $\varepsilon_A = \varepsilon_B = \varepsilon$ the gentle-measurement bound gives $\delta \approx 4\sqrt{\varepsilon}$, while N independent rounds resolve $\delta \sim N^{-1/2}$. The relevant smoothing is therefore $\varepsilon \sim \delta^2/16 \sim 1/N$, not $N^{-1/2}$.

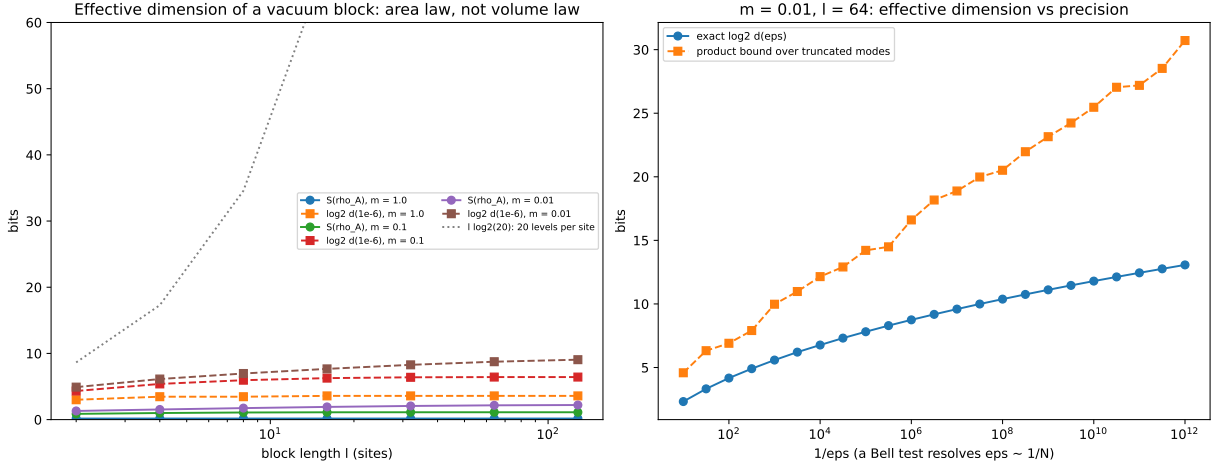


Figure 1: *Left*: entanglement entropy $S(\rho_A)$ (circles) and $\log_2 d_{\rho_A}(10^{-6})$ (squares) of a block of l sites in the vacuum of the harmonic chain for $m = 1, 0.1, 0.01$, compared with the naive count $l \log_2 20$. *Right*: $\log_2 d_{\rho_A}(\epsilon)$ for $m = 0.01$, $l = 64$, together with the product bound of Proposition 4.2.

In QFT the reduced state on a local algebra is not a density operator, since the algebra is of type III. The split property supplies one: the vacuum restricted to the type I factor $\mathcal{N}$ of Theorem 1 is normal and has a density operator. Proposition 4.1 applies to it, and the effective dimension depends on the size of the gap between the regions. For a lattice regularisation the block is itself a type I system.

Vacuum of a lattice free field. Consider the harmonic chain $H = \frac{1}{2} \sum_i [\pi_i^2 + m^2 \phi_i^2 + (\phi_{i+1} - \phi_i)^2]$ on a ring of 4000 sites, a lattice free scalar field, in its ground state. A block of l sites is in a Gaussian state, a product of thermal modes with ratios $q_k = (\nu_k - \frac{1}{2})/(\nu_k + \frac{1}{2})$, where ν_k are the symplectic eigenvalues, i.e. the eigenvalues of $\sqrt{X_A P_A}$ formed from the restricted field and momentum correlators [20]. The spectrum of ρ_A is the set of products $\prod_k (1 - q_k) q_k^{n_k}$. We compute $d_{\rho_A}(\epsilon)$ exactly by best-first enumeration of occupation vectors. Each multiset of excitations is generated once, by incrementing only mode indices not smaller than the last one incremented. Modes with $q_k < 10^{-3}\epsilon/l$ are dropped, which changes the captured weight by less than $10^{-3}\epsilon$.

The single-particle entanglement energies $e_k = \ln(1/q_k)$ explain why few modes matter [19]. They form two towers, one for each endpoint of the block, which is the one-dimensional “area”. Within a tower they grow approximately linearly in k , with a spacing set by the mass: 9.24 for $m = 1$, 4.50 for $m = 0.1$ and about 2.9 for $m = 0.01$ at $l = 64$ and 128. Hence q_k decays exponentially along each tower. The number of modes with $q_k > 10^{-3}\epsilon/l$ is 4 and 6 for $m = 1$ at $\epsilon = 10^{-2}$ and 10^{-6} , independent of l , and 10 and 15 for $m = 0.01$ at $l = 128$. It grows like $\ln(1/\epsilon)$ divided by the spacing, and so it is controlled by the boundary and the correlation length, not by the block volume. Towards criticality the spacing decreases slowly, and the count grows correspondingly.

Table 1 and Figure 1 show two features. First, the effective dimension follows the entanglement entropy, an area law, and not the number of degrees of freedom. For $m = 1$ and $m = 0.1$ it saturates once the block exceeds the correlation length. For $m = 0.01$ it grows logarithmically, like the entropy of a nearly critical chain. At $l = 128$ and $\epsilon = 10^{-6}$ it is 9 bits, while counting 20 levels per site would give 553 bits. Second, the dependence on precision is weak. At $m = 0.01$, $l = 64$ ($S = 2.16$ bits), $\log_2 d(\epsilon)$ grows from 2.3 bits at $\epsilon = 10^{-1}$ to 13.1 bits at $\epsilon = 10^{-12}$. The local slope $d \log_2 d / d \log_2(1/\epsilon)$ falls from 0.60 to 0.25 at $\epsilon \sim 10^{-6}$ and to 0.19 at $\epsilon \sim 10^{-12}$.

Table 1: Entanglement entropy and effective dimension $\log_2 d_{\rho_A}(\varepsilon)$, in bits, for blocks of the harmonic-chain vacuum.

m	l	S	$\varepsilon = 10^{-2}$	10^{-4}	10^{-6}
1	128	0.161	1.00	2.58	3.58
0.1	8	1.075	2.58	4.46	5.95
0.1	128	1.111	2.58	4.86	6.43
0.01	8	1.737	3.32	5.39	6.97
0.01	32	2.062	3.91	6.43	8.28
0.01	128	2.204	4.25	6.99	9.05

Proposition 4.2. Let $\rho = \bigotimes_{k=1}^K \rho_{q_k}$ be a product of thermal modes with spectra $(1 - q_k)q_k^n$. Then

$$d_{\rho}(\varepsilon) \leq \prod_{k=1}^K \left\lceil \frac{\ln(K/\varepsilon)}{\ln(1/q_k)} \right\rceil = O((\ln 1/\varepsilon)^K). \quad (3)$$

Proof. Keep occupations $0, \dots, n_k - 1$ of mode k with $n_k = \lceil \ln(K/\varepsilon) / \ln(1/q_k) \rceil$. The discarded weight of mode k is $q_k^{n_k} \leq \varepsilon/K$, so by the union bound the projector onto the kept product subspace captures weight at least $1 - \varepsilon$. Its rank is $\prod_k n_k$. $\square$

For a block of a regularised field the number of relevant modes is finite, so the certifiable dimension grows polylogarithmically in the precision. By Proposition 4.1, N rounds resolve $\delta \sim N^{-1/2}$, i.e. $\varepsilon \sim 1/N$, so the certifiable number of bits grows only as $O(K \log \log N)$. The bound is loose by a factor of about two in our example (Figure 1, right), but it captures the shape of the growth.

5 Discussion

The companion paper [6] left one way in which a finite experiment could certify an actual infinity: a Bell violation in the gap between C_{qa} and C_{qc} . For local measurements in relativistic QFT with the split property or hyperfinite local algebras, this gap is unreachable (Corollary 3.1). The same holds for the observer algebra of semiclassical de Sitter gravity (Proposition 3.1). Within these frameworks every Bell correlation is approximable by finite-dimensional ones. The approximating dimension is set by the smooth max-entropy of the local state. For the vacuum it follows an area law and grows only polylogarithmically with the precision.

Three caveats bound these conclusions. First, they concern local measurements in the sense of algebraic QFT. Measurements that are not localised, or theories without the split property, for instance with too many local degrees of freedom, are not covered. Second, the hyperfiniteness theorem of [11] assumes a scaling limit, and Proposition 3.1 assumes injectivity of the static-patch QFT algebra. Third, Proposition 4.2 is a statement about regularised blocks. In the continuum the effective dimension depends on the size of the split gap, which acts as a cutoff.

Together with [6], the result is a coherent picture. Physics routinely uses infinite-dimensional Hilbert spaces and type III algebras. Yet the correlations that local measurements can produce in these frameworks are reproduced, to any finite precision, by finite-dimensional models whose size is controlled by entanglement entropy. In this sense, the actual infinity of quantum field theory is not empirically accessible through Bell experiments.

Data and code availability

The script `qft_effective_dimension.py`, the data (CSV), the figure and the console output are available at <https://github.com/Ruqing1963/qft-tsirelson-loop-hole> and archived under DOI 10.5281/zenodo.23202937.

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